Correcting Length Bias

Pivoted Document Length Normalization
First Project Report — Proposal

Group number: 15

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1. Primary Objective

This project will study how document length affects search rankings. We will build a small search system and compare standard cosine-normalized TF–IDF with pivoted document length normalization. We will then test how both methods respond when documents are made longer by adding unrelated text or repeating query terms.

Our main question is: **Does pivoted normalization improve retrieval quality, and does it make rankings more resistant to artificial text manipulation?** These are separate questions: a method that improves ordinary search results may still be vulnerable to keyword repetition.

The project will implement the pivoting idea from Singhal, Buckley, and Mitra's 1996 paper. We will document our implementation choices and differences from the original experiments. This proposal describes planned work; it does not claim completed experiments or results.

2. Evaluation Criteria

We will evaluate both methods using the same documents, queries, and relevance labels.

- **Precision@10:** the fraction of the first ten results that are relevant.
- **Recall@10:** the fraction of all relevant documents found in the first ten results.
- **F1@10:** the harmonic mean of Precision@10 and Recall@10.
- **Mean Average Precision (MAP):** a measure of how consistently relevant documents appear early across the full rankings.
- **Attack effects:** the change in an edited document's rank, the percentage of attacks that move an irrelevant document into the top ten, and the resulting change in retrieval quality.

Queries with no relevant documents will be reported separately and excluded from metric averages. Attack success will mean that an edited, originally irrelevant document enters the top ten; we will report the percentage of successful trials at each attack level.

We will report averages and individual query results. We will also divide the collection into three length groups to examine whether the retrieved results reflect the lengths of relevant documents. These groups will be relative to our collection, since every selected article will have at least 1,000 words. Equal representation of all lengths will not be assumed to be ideal.

3. Datasets Under Consideration

Primary dataset: EPC's *The Fine Print*. We will use 100 articles from the English Press Club's website: https://epcbits.com/, each containing **at least 1,000 words**. The articles have been downloaded with their titles, full body text, source URLs, publication dates, and word counts. They are saved as a CSV file, a JSON file, and individual text files for easy access.

Availability check. On 21 September 2026, we checked all 1,876 posts through the website's public article API. Of these, 153 had at least 1,000 words. We selected the 100 most recent qualifying articles and verified that their IDs, URLs, and extracted texts were distinct. They span **29 November 2020–4 September 2026**, with **1,014–4,343 words per article** and an average of about **1,709 words**. Length is measured from extracted body text before stopword removal, excluding the separately stored title. The final corpus will be reviewed for text quality and topic coverage. Reaching 100 requires archived articles; this selection would not meet a requirement restricting publication to the previous week.

LLM-assisted relevance labeling. We will create about 8–10 queries, each with a short description of the information needed. For every query–article pair, an LLM will receive the query, the article title and full text, and a fixed relevance guideline. It will assign a binary label (1 for relevant, 0 for irrelevant), with a short reason, a supporting excerpt where applicable, and a flag for uncertain cases. Relevance will require substantively addressing the information need, rather than merely mentioning query words.

Group members will review all positive labels and uncertain cases, plus a random sample of negative labels for each query. Incorrect labels will be corrected, disagreements resolved through discussion, and the review expanded if the negative sample reveals frequent errors. We will record the model version, prompt, and number of judgments reviewed. This process will be reported as LLM-generated relevance judgments with human verification; unreviewed labels may still contain errors.

The LLM and reviewers will not see retrieval scores or rankings during labeling. The final labels will be frozen before comparing the methods and used for both systems. Evaluation code will calculate precision, recall, F1, and MAP from these labels; the LLM will not calculate the metrics.

Secondary dataset: Cranfield. Cranfield provides 1,400 scientific abstracts, 225 queries, and relevance judgments. We will use it to check the implementation and choose a slope setting before evaluating the custom dataset. It is available through https://ir-datasets.com/cranfield.html. It will serve as a development dataset; it will not automatically replace a required custom collection.

4. Related Research Papers

Pivoted Document Length Normalization — Amit Singhal, Chris Buckley, and Mandar Mitra (SIGIR, 1996).
Publication link · Open paper

The paper shows that normalization can produce a mismatch between the lengths of retrieved documents and those of relevant documents. It proposes adjusting the normalization factor around a pivot and reports improved retrieval performance on TREC collections. This is the main method we will implement. The paper contains several variants; our core comparison will focus on pivoted cosine normalization.

Improving Term Frequency Normalization for Multi-topical Documents, and Application to Language Modeling Approaches — Seung-Hoon Na, In-Su Kang, and Jong-Hyeok Lee (ECIR, 2008).
https://arxiv.org/abs/1502.02277

This paper distinguishes documents that are long because they repeat a topic from documents that cover several topics. It modifies language-modeling approaches and reports improvements in retrieval effectiveness. This motivates testing keyword repetition and unrelated padding separately in our project.

Two-Stage Document Length Normalization for Information Retrieval — Seung-Hoon Na (ACM TOIS, 2015).
https://arxiv.org/abs/1502.04331

This paper treats repetition and topic coverage in separate normalization stages. Experiments report modest but statistically significant improvements over standard retrieval models. It will help us interpret cases where a single length adjustment helps some long articles but harms others.

Together, these papers establish the baseline method and motivate our extension: comparing ordinary retrieval performance with performance under deliberate text manipulation.

5. Proposed Approach and Architectural Details

The system will follow a simple pipeline:

Articles → text cleaning → shared TF–IDF index → cosine and pivoted ranking → evaluation

Step 1: Prepare the text. We will lowercase and tokenize articles and queries, remove a fixed stopword list, and use identical processing for both methods. The index and scoring functions will be implemented in Python, using NumPy where useful, without a prebuilt TF–IDF vectorizer or retrieval wrapper.

Step 2: Build the baseline. Let $f_{t,d}$ be the number of occurrences of term t in document d, N the number of documents, and df_t the number containing t. We will use logarithmic term frequency and inverse document frequency:

$$TF(t,d) = \begin{cases} 1 + \ln f_{t,d}, & f_{t,d} > 0, \\ 0, & f_{t,d} = 0, \end{cases} \qquad IDF(t) = \ln\left(\frac{N}{df_t}\right).$$

The TF–IDF weight is $w_{t,d} = TF(t,d)\, IDF(t)$. Query weights $w_{t,q}$ use the same rule with query term counts and the collection's IDF. All sums below run over the indexed vocabulary; query terms absent from it will be ignored.

The vector norm measures the overall size of a document's weights. Cosine similarity divides the query–document dot product by both vector norms:

$$n_d = \sqrt{\sum_t w_{t,d}^2}, \qquad n_q = \sqrt{\sum_t w_{t,q}^2}, \qquad S_{\cos}(q,d) = \frac{\sum_t w_{t,q} w_{t,d}}{n_q n_d}.$$

Documents will be ranked by decreasing score. A document or query with a zero norm will receive a score of zero to avoid division by zero.

Step 3: Add pivoted normalization. Let D_+ be the set of documents with nonzero vector norms. We will compute their mean norm p and replace each document's normalization factor with n'_d:

$$p = \frac{1}{|D_+|} \sum_{d \in D_+} n_d, \qquad n'_d = (1-s)p + s n_d, \qquad 0 \le s \le 1.$$

The resulting retrieval score is:

$$S_{\text{piv}}(q,d) = \frac{\sum_t w_{t,q} w_{t,d}}{n_q((1-s)p + s n_d)}.$$

Here, s controls the adjustment, and p is calculated from the clean collection. The pivot must use the same scale as the original norm; we will not substitute average token count into this formula. All other scoring components, including zero-vector handling, remain identical. Setting $s = 1$ reproduces cosine, while $s = 0$ gives every nonzero document the same normalization factor. Intermediate

values reduce the penalty on documents with above-average norms and increase it on those with below-average norms.

We will compare slopes from 0 to 1 in steps of 0.1 on a fixed development split of Cranfield queries, using MAP to select one. We will check the selected setting on held-out Cranfield queries, then freeze it before evaluating EPC. This avoids choosing a setting merely because it gives the best final test result. Our symmetric TF–IDF weighting is an explicit adaptation of the original paper's setup.

Step 4: Test text manipulation. For each query, we will select non-relevant documents that initially lie outside the top ten under both methods. We will edit the same targets for both systems, one document per trial:

- **Keyword repetition:** append the query terms 5, 10, 25, or 50 times.
- **Unrelated padding:** add text equal to 25%, 50%, 100%, or 200% of the original document length, excluding query terms and substantive answers.
- **Combined attack:** add both repeated query terms and unrelated text.

We will retain the original documents and labels. For the main experiment, collection IDF and the pivot will remain fixed while each edited document's weights and norm are recomputed. This isolates the effect of the edit. We will record actual added word counts and compare ranking changes at each attack level.

Step 5: Validate and analyze. Small examples with manually calculated answers will check scoring and metrics. We will then run the corpus experiments and examine queries where the two methods disagree most. A benchmark score that looks plausible will not, by itself, count as proof of correctness.

6. Key Innovations

- Testing whether an established length-normalization method also resists deliberate text manipulation.
- Separating keyword repetition from unrelated padding to understand their different effects.
- Comparing both methods on the same custom corpus, with LLM-assisted relevance labels, human review, and repeatable experiments.

These are extensions within this project, rather than claims of a new normalization formula. A result showing no improvement, or weaker attack resistance, will still be informative.

7. Stretch Objectives

If time permits, we will implement BM25 as an additional comparison, repeat attacks while recomputing collection-wide statistics, and evaluate another dataset or pivoted-normalization variant. These extensions will follow completion of the main comparison.

8. Key Milestones and Tentative Timeline

The schedule is relative to project commencement and can be adjusted to the course deadline.

Period	Planned work	Deliverable
Week 1	Review papers and finalize dataset selection	Literature summary and collection plan
Week 2	Review articles, write queries, and verify LLM labels	Clean corpus, queries, and labels
Week 3	Implement both methods and validate on Cranfield	Working retrieval system and fixed slope

Period	Planned work	Deliverable
Week 4	Run ordinary retrieval and attack experiments	Results tables and plots
Week 5	Analyze findings and complete documentation	Final report and reproducible code

9. Additional Information

The main limitations are the small dataset, few queries, subjective relevance labels, possible residual LLM labeling errors, and artificial attacks. Articles from a single campus source may also share vocabulary and writing style. The 1,000-word minimum excludes short articles, so our conclusions will concern length variation among longer documents. We will report these limitations and avoid assuming that findings generalize to all search collections.

Final project outputs will include the article manifest, queries, relevance labels, implementation, experiment results, and instructions for reproducing them. Dataset collection will retain source links and follow applicable access and reuse conditions.